ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PAPER 2 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET second paper (practical, 100 marks): an mBlock APS points calculator, a C++ electricity bill with a block tariff and VAT, and a C++ class-marks program with arrays, each with an IPO chart.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use mBlock for QUESTION 1 and a C++ development environment (such as Geany or Code::Blocks) for QUESTIONS 2 and 3. If you do not have a computer, write the programs neatly in your answer book.
3. Save each program with the question number and your name or examination number, and add your name or examination number as a comment on the first line of each C++ program.
4. Save your work regularly.
5. Programs are not marked strictly on syntax.

QUESTION 1

Grade 12 learners want to know their Admission Point Score (APS). Create an mBlock app in which the sprite asks for the percentage of each of 6 subjects, changes each percentage into an achievement level, says the level, adds it to the APS and, at the end, says the total APS.

Percentage	**80–100**	**70–79**	**60–69**	**50–59**	**40–49**	**30–39**	**0–29**
Level	7	6	5	4	3	2	1

INPUT	PROCESS	OUTPUT
Percentage of each subject	Declare all variables and set them to 0 Repeat 6 times: Ask for the percentage Use if … else to find the level from the table Say the level aps = aps + level Say the total APS	Level of each subject Total APS

Follow the instructions below to create the mBlock application:

1.1 Open mBlock, create a new project and name it Question1_YourName. (1)

1.2 Choose a sprite to act as a career adviser. (1)

1.3 Choose a backdrop that shows a school or an office. (1)

1.4 Declare all the variables and give them their starting values. (6)

1.5 Use a loop that repeats for 6 subjects. (2)

1.6 Ask for the percentage of a subject and store the answer. (2)

1.7 Use if … then … else blocks to change the percentage into a level. (10)

1.8 Add the level to the APS. (2)

1.9 Display the level of each subject. (2)

1.10 Display the total APS at the end. (2)

1.11 Test the program with the percentages 85, 72, 64, 55, 48 and 39, and write down the levels and the APS that it displays. (3)

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QUESTION 2

A municipality charges for electricity with a block tariff: the first 350 kWh in a month cost R2,50 per kWh and every kWh above 350 costs R3,10. VAT of 15% is then added to the amount. Write a C++ program that works out a household's monthly bill.

INPUT	PROCESS	OUTPUT
Units used (kWh)	Declare all variables and set them to 0 Ask for the units used; if the units are negative, show "Invalid units" If units ≤ 350: amount = units × 2.50 Else: amount = 350 × 2.50 + (units − 350) × 3.10 vat = amount × 0.15 total = amount + vat	Amount before VAT VAT Total to pay

2.1 Create a new C++ source file called Question2_YourName.cpp. (1)

2.2 Include the necessary header(s) and use the std namespace. (2)

2.3 Define the main function and declare all the variables, setting them to 0. (4)

2.4 Ask for the units used and store them; display "Invalid units" and end the program if the value is negative. (4)

2.5 Use if … else to calculate the amount before VAT with the block tariff. (8)

2.6 Calculate the VAT and the total. (3)

2.7 Display the amount before VAT, the VAT and the total, with "R" in front of each amount. (3)

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QUESTION 3

A lecturer wants a C++ program to analyse a class test. The program asks how many students wrote (from 1 to 40), stores each student's mark (0 to 100) in an array, and then shows the class average, the highest and lowest marks and the number of students who passed (50 or more).

INPUT	PROCESS	OUTPUT
Number of students Mark of each student	Declare all variables Ask for the number of students until it is from 1 to 40 Ask for each mark until it is from 0 to 100, and store it in an array Add the marks and work out the average Find the highest and lowest marks Count the marks of 50 or more	Average Highest mark Lowest mark Number passed

The following steps show how the program should be used:

```
How many students wrote (1-40)? 4
Mark for student 1: 67
Mark for student 2: 120
Mark must be from 0 to 100.
Mark for student 2: 45
Mark for student 3: 81
Mark for student 4: 50
Class average: 60.75
Highest mark: 81
Lowest mark: 45
Students who passed: 3
```

3.1	Create a new C++ source file called Question3_YourName.cpp.	(1)
3.2	Include the necessary header(s) and use the std namespace.	(2)
3.3	Define the main function and declare the variables.	(4)
3.4	Ask for the number of students, and ask again until the number is from 1 to 40.	(4)
3.5	Declare the array that stores the marks.	(3)
3.6	Use a loop to ask for each mark, check that it is from 0 to 100 (ask again if it is not) and store it in the array.	(8)
3.7	Calculate the total and the average of the marks.	(4)
3.8	Find the highest and the lowest marks.	(8)
3.9	Count the number of students who passed.	(4)
3.10	Display the average, the highest mark, the lowest mark and the number of passes.	(5)

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TOTAL: 100

END OF QUESTION PAPER